

Menu Recommendation System: Group-Aware Hybrid Recommendation for Food Suggestion and Automatic Combo Generation

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Abstract

Digital restaurant platforms generate a large volume of behavioral data from search logs, menu views, carts, orders, ratings, and repeated purchases. However, many menu systems still display static categories or manually configured promotions, making it difficult to personalize recommendations or create effective food combos. This paper presents CrownDine, a menu recommendation system that applies association rule mining with the Apriori algorithm and collaborative filtering to recommend dishes and automatically construct combos based on user behavior. The main research focus is a group-aware hybrid recommendation model for dine-in contexts, where customers commonly make decisions as a table group rather than as isolated individuals. The proposed model reuses the individual hybrid score generated from Apriori, collaborative filtering, popularity, and contextual signals, then extends it into a consensus-aware group score that rewards average preference while penalizing disagreement among group members. A group-aware combo generation mechanism is also introduced to ensure that a suggested bundle covers the preferences of most members at the table. The system design includes transaction preprocessing, frequent itemset mining, rating and interaction matrix construction, group session modeling, recommendation scoring, combo validation, and evaluation using support, confidence, lift, precision, recall, individual loss, group score, and ranking quality metrics.

Keywords: Menu Recommendation, Group Recommender System, Apriori Algorithm, Association Rule Mining, Collaborative Filtering, Food Combo Generation, Restaurant Analytics

1 Introduction

Restaurant customers increasingly interact with digital menus through mobile applications, QR-code ordering systems, and online food delivery platforms. These interactions produce valuable behavioral signals, including dish views, search keywords, cart additions, order histories, ratings, time-of-day preferences, and repeated purchases. When properly analyzed, such data can help restaurants recommend relevant dishes, suggest complementary items, and create attractive combos that match actual customer behavior rather than relying only on manual menu engineering.

Traditional menu systems often organize dishes by static categories such as appetizers, main courses, drinks, and desserts. Although this structure is easy to browse, it does not adapt to individual preferences or reveal hidden purchasing patterns. For example, customers who order pasta may frequently add red wine, while customers who choose fried chicken may often purchase soft drinks or specific side dishes. If these patterns are discovered automatically, the system can recommend add-ons and generate combo packages that improve convenience for customers and increase average order value for restaurants.

This paper focuses on the topic **Menu Recommendation System: Group-Aware Hybrid Recommendation for Food Suggestion and Automatic Combo Generation Based on User Behavior**. The base system combines two complementary recommendation techniques and extends them into a dine-in group recommendation layer:

1. **Apriori Association Rule Mining:** Discovering frequent itemsets and co-purchase rules from transaction data to identify dishes that are commonly ordered together.
2. **Collaborative Filtering:** Using user-item interaction patterns to recommend dishes based on similarities between users or similarities between menu items.
3. **Hybrid Recommendation Ranking:** Combining association-rule confidence, lift, user preference scores, item popularity, and contextual signals to rank dish suggestions.
4. **Automatic Combo Generation:** Creating dish bundles from frequent itemsets and validating them using business constraints such as category balance, price range, inventory availability, and profit margin.
5. **Group-Aware Recommendation:** Extending the individual score $\text{Score}(u, i)$ into a table-level score $\text{Score}(G, i)$ for dine-in groups by balancing average preference and preference disagreement.

The key research gap addressed in this work is that most menu recommendation models assume that the recommendation receiver is a single user. This assumption is suitable for individual e-commerce or food-delivery scenarios, but it does not fully represent dine-in restaurant behavior, where a reservation or table session usually involves couples, families, friends, or business groups. CrownDine already contains reservation and restaurant-table information, so the proposed contribution focuses on adding a disciplined aggregation layer above the existing recommendation core instead of rebuilding the Apriori or collaborative-filtering modules.

The remainder of this paper is organized as follows. Section 2 reviews related studies on menu recommendation, association rule mining, collaborative filtering, and food combo

design. Section 3 presents the proposed methodology and system architecture. Section 4 discusses expected benefits and limitations, and Section 5 concludes the paper.

2 Related Work

The design of a menu recommendation system is closely related to research in consumer behavior, recommender systems, and restaurant analytics. This section summarizes the main foundations relevant to association-rule-based suggestions and collaborative filtering.

2.1 Digital Menu Recommendation and Consumer Behavior

Digital food ordering platforms influence customer decisions by changing how menu information is presented, filtered, and promoted. Prior studies on restaurant experiences and electronic word-of-mouth show that information quality, credibility, and usefulness affect customer choice intentions [2]. In online ordering environments, perceived convenience, price transparency, and delivery information also shape customer behavior [3]. These findings suggest that recommendation systems should not only display popular dishes but also provide relevant, timely, and trustworthy suggestions that reduce search effort.

Customer satisfaction in restaurants is influenced by food quality, perceived value, and service experience [4, 5]. A recommendation system can support these factors by helping customers find dishes that match their preferences, budget, and meal context. In addition, digital nudging research shows that interface-level interventions can influence food choices without reducing satisfaction [1]. Therefore, recommendations and combo offers should be designed as helpful decision support rather than intrusive advertisements.

2.2 Association Rule Mining for Food Pairing

Association rule mining is widely used to discover relationships among items in transaction databases. In a restaurant context, each order can be treated as a basket containing one or more dishes. The Apriori algorithm identifies frequent itemsets by repeatedly expanding candidate sets whose support exceeds a minimum threshold. From these itemsets, the system can generate rules such as $X \Rightarrow Y$, meaning that customers who order itemset X also tend to order itemset Y .

Association rules are particularly suitable for combo generation because they are interpretable and directly connected to purchasing behavior. Support indicates how common a combination is, confidence measures the probability of buying the consequent given the antecedent, and lift shows whether the relationship is stronger than random co-occurrence. These measures allow restaurant managers to understand why a combo is recommended and to avoid arbitrary bundling.

2.3 Collaborative Filtering for Personalized Dish Suggestions

Collaborative filtering recommends items by learning from interactions between users and items. User-based collaborative filtering identifies customers with similar ordering histories and recommends dishes liked by similar users. Item-based collaborative filtering identifies dishes that are frequently liked or purchased by similar customer groups.

Compared with association rules, collaborative filtering is more personalized because it can recommend items even when they are not part of the same transaction basket.

For menu systems, collaborative filtering can use explicit feedback such as ratings and likes, or implicit feedback such as views, cart additions, purchase frequency, and repeat orders. Because many restaurant systems have sparse rating data, implicit feedback is often more practical. A hybrid design can use Apriori to discover reliable item combinations and collaborative filtering to adapt those combinations to each customer’s historical preferences.

2.4 Automatic Combo Design and Business Constraints

Food combos must satisfy both customer preference and restaurant business constraints. A high-quality combo should include complementary dishes, maintain a reasonable price, avoid redundant categories, and remain feasible under inventory conditions. For example, a lunch combo may include one main dish, one side dish, and one drink, while a family combo may include multiple main dishes and shared appetizers.

Automatic combo generation therefore requires more than mining frequent itemsets. The system must filter candidate combos by category rules, discount policy, preparation feasibility, profit margin, and stock availability. Combining data mining with business constraints allows the restaurant to create practical combos that are both behavior-driven and operationally realistic.

3 Methodology

The proposed system transforms user behavior and transaction records into personalized menu recommendations and automatically generated combos. The methodology consists of data preprocessing, Apriori rule mining, collaborative filtering, hybrid ranking, combo construction, and evaluation.

3.1 Data Collection and Preprocessing

Let $U = \{u_1, u_2, \dots, u_m\}$ be the set of users and $I = \{i_1, i_2, \dots, i_n\}$ be the set of menu items. The system collects behavioral events such as menu views, searches, cart additions, purchases, ratings, and repeat orders. Each order transaction is represented as a basket:

$$T_k = \{i_1, i_2, \dots, i_l\}, \quad T_k \subseteq I. \quad (1)$$

Before model training, raw data are cleaned by removing cancelled orders, merging duplicate item identifiers, normalizing item categories, and filtering inactive menu items. User–item interactions are converted into an implicit preference score:

$$r_{u,i} = w_v V_{u,i} + w_c C_{u,i} + w_p P_{u,i} + w_l L_{u,i}, \quad (2)$$

where $V_{u,i}$ is the number of views, $C_{u,i}$ is the number of cart additions, $P_{u,i}$ is the number of purchases, $L_{u,i}$ is the rating or like signal, and w_v, w_c, w_p, w_l are behavior weights.

3.2 Apriori Association Rule Mining

The Apriori module discovers frequent itemsets from historical order baskets. For an association rule $X \Rightarrow Y$, where $X, Y \subseteq I$ and $X \cap Y = \emptyset$, support, confidence, and lift are defined as:

$$\text{Support}(X \Rightarrow Y) = \frac{|\{T_k \mid X \cup Y \subseteq T_k\}|}{|D|}, \quad (3)$$

$$\text{Confidence}(X \Rightarrow Y) = \frac{|\{T_k \mid X \cup Y \subseteq T_k\}|}{|\{T_k \mid X \subseteq T_k\}|}, \quad (4)$$

$$\text{Lift}(X \Rightarrow Y) = \frac{\text{Confidence}(X \Rightarrow Y)}{\text{Support}(Y)}. \quad (5)$$

Rules are retained when they satisfy minimum support, confidence, and lift thresholds. For example, if customers who order grilled beef frequently order salad and fruit juice, the system can generate a rule that recommends those add-ons when grilled beef is selected. The Apriori output is also used as the candidate pool for automatic combo construction.

3.3 Collaborative Filtering Model

The collaborative filtering module builds a user-item interaction matrix $R \in \mathbb{R}^{m \times n}$, where $r_{u,i}$ represents the preference score of user u for item i . In item-based collaborative filtering, the similarity between two dishes i and j is computed using cosine similarity:

$$\text{sim}(i, j) = \frac{\sum_{u \in U} r_{u,i} r_{u,j}}{\sqrt{\sum_{u \in U} r_{u,i}^2} \sqrt{\sum_{u \in U} r_{u,j}^2}}. \quad (6)$$

The predicted preference of user u for an unseen dish i is calculated as:

$$\hat{r}_{u,i} = \frac{\sum_{j \in N(i)} \text{sim}(i, j) r_{u,j}}{\sum_{j \in N(i)} |\text{sim}(i, j)|}, \quad (7)$$

where $N(i)$ is the set of items similar to i that user u has interacted with. Items with high predicted scores are recommended as personalized suggestions.

3.4 Hybrid Ranking and Recommendation Output

The final recommendation score combines collaborative filtering predictions with association-rule evidence and popularity signals:

$$\text{Score}(u, i) = \alpha \hat{r}_{u,i} + \beta AR(i \mid B_u) + \gamma Pop(i) + \delta Ctx(u, i), \quad (8)$$

where $AR(i \mid B_u)$ is the association-rule score for item i given the user's current basket B_u , $Pop(i)$ is normalized item popularity, $Ctx(u, i)$ represents contextual compatibility such as meal time or category preference, and $\alpha, \beta, \gamma, \delta$ are tunable weights.

The system produces two outputs. The first output is a ranked list of recommended dishes for each user. The second output is a set of suggested add-ons based on the user's current basket, such as drinks, side dishes, or desserts that frequently appear with selected items.

3.5 Automatic Combo Generation

Combo generation starts from frequent itemsets and high-lift association rules. A candidate combo C is defined as:

$$C = \{i_1, i_2, \dots, i_q\}, \quad q \geq 2. \quad (9)$$

Each candidate combo is validated using category and business constraints:

$$\text{CategoryBalance}(C) = 1, \quad (10)$$

$$P_{\min} \leq \text{Price}(C) \leq P_{\max}, \quad (11)$$

$$\text{Margin}(C) \geq M_{\min}, \quad (12)$$

$$\text{Stock}(i) > 0 \quad \forall i \in C. \quad (13)$$

The combo score is computed as:

$$\text{ComboScore}(C) = \lambda_1 \text{Support}(C) + \lambda_2 \text{Lift}(C) + \lambda_3 \text{Margin}(C) + \lambda_4 \text{Preference}(u, C). \quad (14)$$

Combos with high scores are presented to customers as dynamic bundles, while restaurant administrators can review suggested combos before publishing them as official promotions.

3.6 Group-Aware Hybrid Recommendation for Dine-In Context

The original hybrid score $\text{Score}(u, i)$ evaluates how suitable item i is for one user u . In dine-in restaurants, however, the practical decision unit is often a group sharing the same table. Let

$$G = \{u_1, u_2, \dots, u_k\} \subseteq U \quad (15)$$

be a group of customers associated with one restaurant table and one reservation session. The group-aware recommendation problem is to compute $\text{Score}(G, i)$, which represents the suitability of item i for the entire table, while avoiding dishes that are highly preferred by some members but strongly disliked by others.

To support this model, CrownDine introduces a lightweight dining group session layer. A *DiningGroupSession* represents an active group meal associated with a *Reservation* and a *RestaurantTable*. A *GroupMember* links a registered user, or optionally a guest customer, to that session. When an organizer reserves a table or scans the table QR code, the system creates a group session. Other members join by scanning the same table QR code. For registered members, the system uses their personal hybrid scores. For guest members without historical interactions, the system uses normalized popularity $\text{Pop}(i)$ as a fallback preference estimate.

The extension deliberately does not replace the existing Apriori or collaborative-filtering pipeline. For every member $u_j \in G$, the system first computes $\text{Score}(u_j, i)$ using the same individual hybrid equation. The group layer then aggregates these individual scores. A simple average aggregation is defined as:

$$\text{Score}_{\text{avg}}(G, i) = \frac{1}{|G|} \sum_{u_j \in G} \text{Score}(u_j, i). \quad (16)$$

Although this approach captures the central tendency of group preference, it may still recommend divisive dishes. Therefore, this paper proposes a consensus-aware aggregation strategy that penalizes disagreement:

$$\text{Disagreement}(G, i) = \sqrt{\frac{1}{|G|} \sum_{u_j \in G} (\text{Score}(u_j, i) - \text{Score}_{avg}(G, i))^2}, \quad (17)$$

$$\text{Score}_{CA}(G, i) = \text{Score}_{avg}(G, i) - \mu \cdot \text{Disagreement}(G, i), \quad (18)$$

where $\mu \geq 0$ controls how strongly the system prioritizes group consensus. When $\mu = 0$, the model becomes average aggregation. As μ increases, the model becomes more conservative and avoids items with high variance in member preferences. This design reflects the dine-in objective that a shared recommendation should satisfy the table as a whole rather than maximizing only one customer's preference.

Group-aware combo generation further extends the original combo score. A good table-level combo should not only have strong support, lift, and margin, but should also include at least one attractive item for most group members. This condition is captured by preference coverage:

$$\text{Coverage}(C, G) = \frac{1}{|G|} |\{u_j \in G : \exists i \in C, \text{Score}(u_j, i) \geq \theta\}|, \quad (19)$$

where θ is the minimum personal preference threshold. The group combo score is then computed as:

$$\text{GroupComboScore}(C, G) = \text{ComboScore}(C) + \varepsilon \cdot \text{Coverage}(C, G), \quad (20)$$

where ε controls the importance of covering the preferences of the whole group. This mechanism formalizes a practical restaurant requirement: a shared combo should be acceptable to most people at the table, not only profitable or frequently purchased.

At the service layer, the proposed extension can be implemented inside the existing Recommendation and Analytics Service by adding group-level operations such as `getGroupRecommendations(sessionId, topK)`, `generateGroupCombo(sessionId)`, and `computeDisagreement(sessionId, itemId)`. Because the individual recommendation core remains unchanged, the integration cost is low and the model can be deployed as an additional aggregation layer above the current recommendation workflow.

3.7 Evaluation Metrics

The recommendation quality is evaluated using ranking and classification metrics. Precision@K measures how many of the top- K recommendations are relevant, while Recall@K measures how many relevant items are retrieved:

$$\text{Precision@K} = \frac{|\text{Recommended@K} \cap \text{Relevant}|}{K}, \quad (21)$$

$$\text{Recall@K} = \frac{|\text{Recommended@K} \cap \text{Relevant}|}{|\text{Relevant}|}. \quad (22)$$

Combo quality is evaluated using support, confidence, lift, average order value improvement, click-through rate, and conversion rate. These metrics help determine whether the system improves both customer experience and restaurant revenue.

For the group-aware extension, two additional metrics are used. Individual loss measures how much each member sacrifices compared with the item that would have been best for that member alone:

$$\text{IndividualLoss}(G) = \frac{1}{|G|} \sum_{u_j \in G} (\text{Score}(u_j, i_j^*) - \text{Score}(u_j, i_{\text{chosen}})), \quad (23)$$

where i_j^* is the highest-scoring item for member u_j and i_{chosen} is the item or combo selected for the group. Lower individual loss means that the group recommendation sacrifices fewer personal preferences. Group score is collected through a post-meal satisfaction survey on a five-point scale and averaged across members. In a pilot evaluation, the system can compare three variants: no group recommendation, average-only aggregation with $\mu = 0$, and consensus-aware aggregation with $\mu > 0$. A minimum of 30–50 group dining sessions per variant is recommended for an initial course-project pilot.

4 Discussion

The proposed system is suitable for restaurants that want to move from static menu presentation to data-driven personalization. Apriori association rules provide transparent evidence for add-on suggestions and combo creation, making them easy for restaurant managers to interpret. Collaborative filtering complements this by adapting recommendations to each user’s previous behavior, which is important when different customers have different tastes, budgets, and ordering habits.

A hybrid approach is especially useful in restaurant environments because recommendation goals are both customer-oriented and business-oriented. Customers benefit from faster dish discovery and more relevant suggestions, while restaurants benefit from increased cross-selling, improved combo design, and better use of transaction data. The group-aware extension strengthens this contribution by aligning the algorithm with the dine-in context of CrownDine. Instead of assuming that every recommendation is consumed by one isolated user, the proposed model treats the table as the decision unit and uses reservation, restaurant-table, and QR-based group-session data to generate recommendations for the whole group.

The consensus-aware score provides a focused research contribution because it does not merely add another signal to the original hybrid formula. It changes the recommendation target from $\text{Score}(u, i)$ to $\text{Score}(G, i)$ and explicitly models preference disagreement, which is a central issue in shared dining. This is more suitable for group meals than a pure average strategy, because an item with a high average score may still be unacceptable if one or two members strongly dislike it. The proposed coverage term for combos follows the same principle by requiring a bundle to satisfy most members at the table.

Several challenges must be addressed. New users and new menu items may suffer from cold-start problems because there is not enough interaction history. Guest members can only be approximated through popularity scores, which may weaken the accuracy of disagreement estimation when a group contains many anonymous customers. Privacy is also important because explanations such as “recommended for a specific member” may reveal personal preferences to other diners; therefore, the interface should prefer anonymous explanations such as “suitable for most members at this table.” For large groups, disagreement may become consistently high, so future work should investigate subgroup clustering for banquets or company meals. Automatically generated combos

must also be reviewed carefully to avoid poor taste combinations, excessive discounts, or operational conflicts in food preparation.

5 Conclusion

This paper presented CrownDine, a menu recommendation system that applies Apriori association rule mining and collaborative filtering to suggest dishes and automatically generate food combos based on user behavior. Apriori discovers interpretable co-purchase patterns from transaction baskets, while collaborative filtering personalizes recommendations using user-item interaction data. The hybrid ranking mechanism combines these signals with popularity, context, and business constraints.

The main deep-dive contribution is the extension from individual recommendation to group-aware dine-in recommendation. By introducing DiningGroupSession, GroupMember, consensus-aware aggregation, disagreement penalization, and group combo coverage, the system can compute $\text{Score}(G, i)$ for customers sharing the same table while reusing the original Apriori and collaborative-filtering core. This makes the contribution both technically focused and practical for restaurant deployment.

The proposed system provides a practical foundation for intelligent digital menus. By recommending relevant dishes and constructing behavior-driven group combos, restaurants can reduce customer decision effort, increase average order value, and improve the effectiveness of promotions. Future development will focus on calibrating the disagreement weight μ , handling guest-heavy groups, protecting preference privacy, applying subgroup clustering for large tables, and evaluating the model through real dine-in A/B testing using individual loss and group satisfaction score.

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